

Aniruddha Chowdhury, Ph.D., EIT

Computational Materials Science | Molecular Simulation | Sustainable Infrastructure Materials

106 S Homer St, Apt 19, Ruston, LA 71270 | 469-493-2577 | ani.nitjsr@gmail.com

[LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

RESEARCH INTERESTS

Multiscale molecular dynamics simulation of polymer-asphalt systems; coarse-grained force field development (MARTINI 3); asphaltene aggregation and oxidative aging; polymer compatibility and phase behavior; recycled-plastic and bio-modified binders.

EDUCATION

Ph.D. in Micro and Nanoscale Systems Louisiana Tech University, Ruston, LA	Mar 2021 – May 2026
M.Sc. in Mechanical Engineering Louisiana Tech University, Ruston, LA	Mar 2021 – Aug 2023
B.Tech. in Mechanical Engineering National Institute of Technology, Jamshedpur, India	Aug 2012 – Jun 2016

CERTIFICATIONS & LICENSURE

• Licensed Engineer Intern, Louisiana (EI.0036277) • P.E. exam passed (Mechanical) • Lean Manufacturing Certification • PMP Training • Six Sigma Green Belt

RESEARCH EXPERIENCE

Graduate Research Assistant, Louisiana Tech University, Ruston, LA	Mar 2021 – May 2026
– Led the computational thrust of a multi-year federally funded project (Louisiana DOTD / Federal Highway Administration, Grant No. 693JJ320C000024) applying molecular dynamics simulation to develop deployable recycled plastic-asphalt binder systems at national scale. – Developed and validated novel MARTINI 3 coarse-grained representations of asphalt binder, enabling microsecond- and micrometer-scale simulation of polymer dissolution and phase separation that was previously inaccessible to atomistic models. – Established a revised molecular model for asphalt oxidative aging and used it to predict polymer compatibility across binder chemistries. – Characterized asphalt-polymer blends experimentally by FTIR spectroscopy to validate simulated mixing and phase behavior. – Built automated analysis pipelines in LAMMPS, Python, MATLAB, JMP, and SQL for large trajectory datasets and released the simulation and analysis codes publicly through GitHub. – Transferred the modeling framework to industry as a high-level screening tool; adopted by an industry engineer (Johns Manville, later Caltrans) in a pilot program incorporating recycled plastic in asphalt binder, and by an academic group in Turkey. – Directed cross-functional research teams and mentored graduate and undergraduate students in molecular simulation and thermochemical data analysis. – Contributed to a NASA EPSCoR collaboration (Louisiana Board of Regents) with Louisiana State University and Tulane University in partnership with NASA Glenn Research Center.	

PEER-REVIEWED JOURNAL PUBLICATIONS

1. **Chowdhury, A.**, Nourian, P., Wasiuddin, N., & Peters, A. (2026). Development of a revised molecular model for asphalt oxidation and its implications for polymer compatibility. *Fuel*.
2. **Chowdhury, A.**, Nourian, P., Wasiuddin, N., & Peters, A. (2024). Investigation of polymer-asphalt compatibility using molecular dynamics simulation. *Journal of Applied Chemistry B*.

Manuscripts Under Review

1. **Chowdhury, A.**, Nourian, P., Wiley, J., & Peters, A. Analysis of polymer-asphalt dissolution dynamics using MARTINI based coarse-grained simulation. *Journal of Molecular Liquids* (final review).
2. Three additional co-authored manuscripts currently under review.

NOTABLE CONFERENCE PRESENTATIONS & PROCEEDINGS

1. **Chowdhury, A.**, Peters, A., Nourian, P., & Wasiuddin, N. (2026, January). Investigating the effect of asphalt-polyethylene compatibility using molecular dynamics simulation. *Transportation Research Board Annual Meeting*, Washington, D.C.
2. **Chowdhury, A.**, Nourian, P., & Wasiuddin, N. (2024, December). Investigation of phase separation behavior of polymer-asphalt mix using molecular dynamics simulation. *Gulf Coast Polymer Conference*, University of Southern Mississippi, Hattiesburg, MS.
3. Peters, A., & **Chowdhury, A.** (2024, January). A coarse-grained model for asphalt binder and polymer mixtures based on the MARTINI force field. *Transportation Research Board Annual Meeting*, Washington, D.C.
4. **Chowdhury, A.**, Peters, A., Nourian, P., & Wasiuddin, N. (2023, November). Investigation of waste plastic-asphalt compatibility using MD simulation. *American Institute of Chemical Engineers Annual Meeting*, Orlando, FL.
5. Peters, A., **Chowdhury, A.**, Rahman, R., Shafe, A., Hossain, R., & Wasiuddin, N. (2021, November). Use of waste plastic in asphalt binder: Theoretically and experimentally justified identification of compatible blends. *American Institute of Chemical Engineers Annual Meeting*, Boston, MA.
6. **Chowdhury, A.** (2023, 2024, 2025). Oral presentations, *Graduate Research Symposium*, Louisiana Tech University, Ruston, LA.

PROFESSIONAL SERVICE & PEER REVIEW

- Invited manuscript reviewer: *RSC Advances*, *The Journal of Physical Chemistry*, *ACS Omega*, *Transportation Research Board*, and *International Journal of Pavement Research and Technology*.
- Mentored graduate and undergraduate researchers on molecular dynamics and polymer chemistry principles.

AWARDS & HONORS

- Outstanding Graduate Student Award, College of Engineering and Science, Louisiana Tech University, 2025.
- 1st Place, Oral Presentation, Graduate Research Symposium, Louisiana Tech University, 2023 and 2024; 3rd Place, 2025.
- Harry J. Gaston, Jr. Memorial College of Engineering and Science Scholarship, 2025.
- Adamantios K. Gorgolis Graduate College of Engineering and Sciences Scholarship, 2025.
- Indian Council for Cultural Relations (ICCR) fully funded merit scholarship, Government of India, for undergraduate study.

TECHNICAL SKILLS

Simulation & Computation: LAMMPS, GROMACS-style coarse-grained workflows (MARTINI 3), VMD, Python, MATLAB, SQL, Linux/HPC job management, automated trajectory analysis.

Characterization: FTIR, SEM with EDS, LSCM, AFM, XPS, TGA; photolithography, wet and dry etching, CVD and thin-film deposition, four-point probe I-V and C-V measurement.

Data, Statistics & Engineering Tools: JMP, Minitab, Power BI, SolidWorks, AutoCAD, Revit, DMAIC, root cause analysis, FMEA.

PROFESSIONAL EXPERIENCE

Process Engineer, Al Razi Chemical Complex Limited, Bangladesh

Jan 2018 – Dec 2020

- Modeled a hydrogen peroxide production process in Aspen Plus for material and energy balance analysis.
- Received a Kaizen Award for raising distillation unit process capability from a Cpk of approximately 1.14 to 1.40 using Six Sigma methods and root cause analysis.
- Supervised 25+ operators, SOPs, and safe chemical handling; tracked KPIs and reported daily production performance to senior engineers.

Assistant Mechanical Engineer, AFTAB Automobiles Limited, Bangladesh

Aug 2016 – Dec 2017

- Validated bus HVAC sub-assemblies through leak, pressure-integrity, and airflow testing, releasing roughly 30 units per month at 100% first-pass yield.
- Supervised quality validation of 3,000+ injection-molded seat assemblies and FRP compression molding of exterior and interior structures; modeled chassis and prototype components in SolidWorks and AutoCAD.

REFERENCES

Dr. Andrew Peters (Doctoral Advisor), Terrell and Henry Families Associate Professor in Engineering, Louisiana Tech University | apeters@latech.edu

Dr. Nazimuddin Wasiuddin, Professor and MCI Endowed Professor, Mississippi State University | wasi@cee.msstate.edu

Dr. Jack Youtcheff, Principal Infrastructure Materials Research Engineer, Federal Highway Administration | jack.youtcheff@dot.gov

Dr. Md Shams Arafat, Transportation Asset Management Engineer, California Department of Transportation (Caltrans) | md.shams.arafat@dot.ca.gov